

The 5-Layer Knowledge Architecture

To keep your files organized, fast, and accessible across local scripts, Obsidian, and LLM repositories, structure your data into these distinct functional tiers:

Tier 1: Raw Source Layer (/raw_pdf/)

- **Format:** .pdf / Raw Docs
- **Role:** The untouchable source of truth. Heavy, unedited reference manuals, research papers, and technical specs.
- **Storage:** Stored in deep local storage or Google Drive. You rarely feed these raw files directly to an LLM due to context weight.

Tier 2: Conceptual Wiki Layer (/wiki_md/)

- **Format:** .md (Markdown)
- **Role:** Consolidated, human-readable summaries and briefs.
- **Target:** Obsidian Vault / LLM Wiki repo.
- **Structure:** Uses strict internal linking ([[links]]) or relative paths) so both you and an LLM can traverse concepts without reading 50-page PDFs.

Tier 3: High-Speed Machine Caches (/recall_jsonl/)

- **Format:** .jsonl (JSON Lines)
- **Role:** Structured, line-delimited metadata optimized for instantaneous parsing by on-device agents and local scripts.
- **Structure:** Each line is a self-contained record:

JSON

```
"id": "doc_001", "topic": "QAIRT_conversion", "summary": "...", "source_pdf": "specs/qairt.pdf"  
"wiki_link": "wiki/qairt.md"
```

Tier 4: Execution & Runtimes (/repos/ & /scripts/)

- **Format:** .sh, .py, codebases
- **Role:** The actual working code, model conversion scripts, shell loops, and active toolchains.
- **Rule:** Scripts should point to Tier 3 (.jsonl) or Tier 2 (.md) for their configuration and reference data rather than hardcoding paths.

Tier 5: The Master "Almanac" / Index (INDEX.jsonl or MAP.md)

- **Format:** Single root file or global database line-map
- **Role:** The glue that bridges Tiers 1 through 4. It maps every raw PDF to its corresponding Markdown wiki summary, its JSONL cache offset, and the specific script repo that executes it.

Why This Pipeline Works

1. **Speed:** When your local agent needs to recall facts, it scans the **Tier 3 JSONL** file in milliseconds without needing to read heavy Markdown files or parse PDFs.
2. **Context Preservation:** When you need deep reasoning in a chat interface or Obsidian, you point the agent to the **Tier 2 Markdown** brief.
3. **Traceability:** If the summary feels incomplete, the **Tier 5 Index** gives you the exact line number and source PDF path to inspect.